MOHAMED SIDDIQ

AI Engineer | GenAI Engineer

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PROFESSIONAL SUMMARY

AI Engineer and 2026 B.Tech graduate in Artificial Intelligence & Data Science, with hands-on experience building end-to-end LLM applications in Python. Skilled in RAG pipelines with vector databases, multi-agent systems and voice AI using LangChain, LlamaIndex, FastAPI, ChromaDB, and OpenAI, Azure OpenAI and Gemini models. Brings 3 years of end-to-end digital marketing experience, independently managing social media, Google Ads and Meta Ads campaigns, and sales lead generation. Seeking an AI Engineer role.

TECHNICAL SKILLS

Languages & APIs: Python, SQL, FastAPI, Flask, REST APIs, Debugging & Troubleshooting

Generative AI: Gen AI, LLMs (OpenAI GPT-4o, o3-mini, Azure OpenAI, Gemini), Prompt Engineering, NLP, Embeddings, Hugging Face

LangChain & Agentic AI: LangChain, LlamaIndex, Agentic AI, AI Agents, Multi-Agent Systems, CrewAI, AutoGen, n8n, Zapier

RAG & Vector Search: RAG Pipelines, Vector Databases (ChromaDB), Hybrid Search, Semantic Search, Data Ingestion Pipelines

AI Models & Data: ML, Deep Learning, TensorFlow (Basics), Scikit-Learn, Pandas, NumPy, Model Evaluation, Performance Optimization

Cloud & DevOps: Docker, Git, GitHub, Azure OpenAI, AWS (Basics), Apache Airflow

Quality & Teamwork: Guardrails, Hallucination Control, Edge-Case Testing, Cross-Functional Collaboration, Problem Solving

PROJECTS

Hybrid Voice RAG AI Assistant

Tech Stack: *Python, Streamlit, OpenAI API (GPT-4o-mini, Whisper, TTS-1-HD), LangChain, ChromaDB, BM25Okapi, Pydub*

Overview: Voice and text assistant that answers questions from users' own uploaded documents, in 7 Indian languages.

- Engineered an end-to-end voice-enabled RAG assistant with a document ingestion and indexing pipeline, supporting real-time spoken and text interaction over uploaded documents.
- Built a high-recall hybrid retrieval engine combining dense semantic search (ChromaDB) and sparse keyword matching (BM25Okapi) via LangChain, so both meaning-based and exact-term queries return relevant context.
- Implemented adaptive multilingual NLP with natural language switching across 7 languages (English, Hindi, Tamil, Telugu, Malayalam, Kannada, Urdu).
- Created a speech pipeline using Whisper for speech-to-text and TTS-1-HD with Pydub for clear, context-aware voice responses.
- Developed a Streamlit interface for document upload, text chat, and voice input/output, orchestrating retrieval, GPT-4o-mini generation, and speech synthesis as one workflow.

Enterprise GenAI RAG Application

Tech Stack: *Python, OpenAI API, LangChain, ChromaDB, FastAPI*

Overview: Conversational platform answering questions grounded in a company's private, unstructured data.

- Designed an enterprise-style GenAI application that turns unstructured data into actionable insights using a RAG pipeline and vector database grounded in private domain knowledge.
- Reduced hallucinations by grounding every LLM response in retrieved domain context, and added real-time response streaming and dynamic prompt orchestration.
- Implemented domain guardrails for secure, scalable use, and exposed the pipeline through FastAPI for integration with enterprise applications.

Multi-Agent Customer Support System

Tech Stack: *Python, CrewAI, FastAPI, Gemini API / OpenAI, SQLite / PostgreSQL, HTML/JS*

Overview: Automated customer support platform where specialised AI agents resolve tickets and escalate hard cases.

- Built a multi-agent customer support platform with 4 role-based agents (Support, Technical, Escalation, QA) to automate end-to-end issue resolution.
- Engineered a routing engine that scores query complexity and priority and transfers high-priority or edge-case tickets to escalation workflows.
- Developed REST API endpoints with persistent SQLite/PostgreSQL storage for full ticket lifecycle management and conversation state retention.
- Built an HTML/JS front end and supported Gemini and OpenAI as interchangeable LLM providers.

AutoGen Customer Support Agent

Tech Stack: *Python, Microsoft AutoGen, OpenAI / Azure OpenAI (GPT-4o, o3-mini), Pandas, FuzzyWuzzy*

Overview: Multi-agent support assistant that escalates complex cases to a stronger reasoning model.

- Built an autonomous support framework with 3-agent collaboration: User Proxy Agent, General Agent, and Senior Agent.
- Implemented dual-tier escalation: GPT-4o for standard queries and o3-mini for complex technical disputes, safety issues, and policy exceptions.
- Integrated FuzzyWuzzy matching with CSV knowledge bases (Pandas) for fast retrieval, and orchestrated multi-turn conversations with GroupChatManager, conversation logs and edge-case test scenarios.

PROFESSIONAL EXPERIENCE

Operative Executive – Digital Marketing | **Evoulth Digital Pvt. Ltd., India** *2022 – 2025*

- Independently managed end-to-end digital marketing, from campaign strategy and creative production to execution and performance tracking.
- Ran lead-generation campaigns on Meta Ads and Google Ads, turning social media traffic into direct sales inquiries and calls while optimizing ROAS, CPC, and CAC.
- Created and edited video content for paid ads and social media, and managed brand social media accounts end to end.
- Improved search rankings and organic traffic through on-page and off-page SEO; tracked results in Excel dashboards.
- Worked with sales and growth teams to turn campaign data into optimization decisions and reduce funnel drop-offs; coordinated email and bulk SMS campaigns.

EDUCATION

Bachelor of Technology, Artificial Intelligence & Data Science *2022 – 2026*
Mohamed Sathak A.J. College of Engineering, Chennai | CGPA: 7.0

CERTIFICATIONS

AI Engineer – Unstop | Business Analytics – IBM | Data Analytics – Novitech Pvt. Ltd. | Digital Business Skills – HCL

LANGUAGES

English | Tamil | Hindi | Urdu